

基础14 C++申请内存的各种方法

① 内存的手动申请与释放

a. malloc / free

```
void * p1 = malloc(512); free(p1);
```

C语言函数 malloc: 只申请固定大小的内存

free: 释放内存

注: free在释放内存时未传入内存大小

所以释放内存至少比申请的内存(512)更大

b. new / delete

```
Test *p2 = new Test();
```

```
void *mem = operator new(sizeof(Test));
```

 申请内存

```
Test *p2 = (Test *) mem;
```

 指针转换

```
p2 → Test::Test();
```

 在内存上构造对象

```
delete p2; p2 → ~Test();
```

 先析构对象

```
operator delete p2;
```

 再释放内存

new与delete为C++函数, 内部有类的构造与析构

c. placement new

```
Test *p3 = new(p1) Test();
```

```
void *mem = operator new(sizeof(Test), p1);
```

 指定地方申请内存

```
Test *p3 = (Test *) mem;
```

```
p3 → Test::Test();
```

 只能delete(p3); 需要调用析构

placement new 使用已分配内存的指针

本身不会再申请额外空间

②完全掌握内存的申请与释放: 重写类内的 operator new/delete

```
class Test{...
```

```
void *operator new(size_t size){
```

```
    return malloc(size);
```

```
void *operator new(size_t size, void *buf){
```

```
    return buf;
```

```
void *operator delete(void *ptr){
```

```
    free(ptr);
```